

Brief Article

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1 V Known in Advance

2 V Realized with Outcomes

Assume: V and c are independent, cost paid by all people listed (not just those who actually donate).

Death rate β ; number of organs per death α ; rate of need θ .

Assume undersupply (overdemand): $\alpha\beta < \theta$. This implies that under both donor and donor + value, only donors will receive organs.

2.1 Donor Only

I donate at cost c , and benefit with probability $\theta \left(\frac{\alpha\beta}{\theta} \right)$, with average benefit $E[V] = \int_0^\infty \overline{G}(v)dv$. Threshold will be $c_d^* = \alpha\beta E[V]$.

Welfare gain from transplantation is number of transplants times average benefit of a transplant, which is $\alpha\beta E[V]F(\alpha\beta E[V])$. **Comes at donation cost...**

Cost to marginal donor, c^* , is smaller than benefit to average recipient, $E[V]$.

2.2 Donor + Value

Define v^* to be solution to $\overline{G}(v^*) = \alpha\beta/\theta$.

I donate at cost c , and benefit with probability $\theta \overline{G}(v^*) = \theta \left(\frac{\alpha\beta}{\theta} \right)$, with average benefit $E[v|v \geq v^*] = v^* + \frac{1}{\overline{G}(v^*)} \int_{v^*}^\infty \overline{G}(v)dv$. Threshold will be $c_{dv}^* = \theta (v^* \overline{G}(v^*) + \int_{v^*}^\infty \overline{G}(v)dv)$, which can be shown to be higher than before, so more donate.

Welfare is number of transplants, $\alpha\beta F(c_{dv}^*)$, times benefit per transplant $E[v|v \geq v^*]$. **Comes at donation cost...**